

Validation of Thin Layer Chromatographic Methods

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13.1 INTRODUCTION

As it is the case with all analytical methods that are assaying compounds, thin-layer chromatographic and high-performance thin-layer chromatographic (TLC/HPTLC) methods must be validated to ensure that they are fit for their intended purpose and, where applicable, meet the strict regulatory requirements for controlled products, such as foods and pharmaceuticals. Validation procedures carried out for other analytical techniques can also be applied to TLC/HPTLC methods to evaluate their performance parameters. In this chapter, the discussion on validation procedures of TLC/HPTLC analytical methods will be based on the validation of methods for pharmaceutical analysis since validation in this area is considered to be quite strict.

Method validation, as observed in the literature, may be categorized into classical [1–5] and alternative [6–8] approaches. The classical approach, which determines a number of method performance parameters, is widely described in the literature and is the most commonly used, whereas the alternative approach was more recently introduced by the *Société Française des Sciences et Techniques Pharmaceutiques* (SFSTP) [6], and

is more frequently applied nowadays. It is based on the use of the so-called accuracy profiles.

13.2 METHOD VALIDATION USING THE CLASSIC APPROACH

The sources of information in the literature describing the classical validation procedures for analytical methods include among many others, the International Conference on Harmonization of Technical Requirements for the Registration of Pharmaceuticals for Human Use (ICH) guidelines [1], the Food and Drug Administration Guidance for Industry [2], the United States Pharmacopeia [3], the Association of Official Analytical Chemists guidelines for method validation for a single laboratory [4], and the Quality Assurance guideline in Analytical Chemistry [5].

According to the above references, method performance parameters to be evaluated are the linearity of the calibration line, precision, trueness, range, limits of detection and quantification, specificity, and robustness. The extent to which a TLC/HPTLC method is validated, i.e., which parameters are evaluated, depends on the intended goals to be achieved and the intended use. For example, a qualitative TLC/HPTLC procedure will be less extensively validated and its validation will be limited to evaluating the specificity and detection limit, while for a quantitative procedure, more parameters, such as linearity of the calibration line, precision, trueness, limits of detection and quantification, and specificity, will be evaluated. When the method is to be used in many different laboratories, it most probably also will be subjected to a robustness test. In a classical validation approach, generally the performance parameters are evaluated individually and the result for each one of them is compared with an acceptance criterion.

13.2.1 Linearity of the Calibration Line

Analysts generally prefer to work with linear equations allowing an easy estimation of the amount of sample ingredient(s). In these equations, a response related to the spot (e.g., peak area after scanning) on the plate is related to its concentration. With TLC/HPTLC methods, the concentration in the standard (sample) is often expressed as the amount (mass) spotted. For these methods, straight lines may be obtained when dilute solutions are used within limited concentration ranges. However, in some cases, due to the saturation effect taking place at the application point on the plate, quadratic relationships are reported to offer acceptable results. For instance, Habte et al. [9] developed and validated a method for the simultaneous separation and determination of lamivudine (LVD) and

zidovudine in pharmaceutical formulations. The calibration data fitted well to the quadratic model with $r^2 = 0.9997$ and 0.9996 for LVD and zidovudine, respectively, as compared to a straight-line model, where the data fitted less well ($r^2 = 0.9776$ and 0.9616 , respectively). The mean assay results obtained using the quadratic models were $99.74 \pm 0.84\%$ and $99.76 \pm 0.88\%$, for LVD and zidovudine, respectively.

Kaale et al. [10] participated in and described the results of an inter-laboratory investigation on the use of HPTLC methods to perform assays of LVD–zidovudine, metronidazole, nevirapine (NVP), and quinine composite samples. A linearity study for each compound was performed at concentration levels covering 50–150% of the target concentration. The calibration data for all compounds fitted well to quadratic models with r^2 values above 0.9980. The assay results obtained using the quadratic models for the participating laboratories for all compounds ranged from 97.0 to 102.7% with the percent relative standard deviations (%RSD) $\leq 3.21\%$.

When linear calibration lines are desired, usually a coefficient of determination (r^2) of at least 0.99 is considered by many authors as an indicator of the linearity of the calibration lines. However, this is not a good criterion [11–14] since it has been demonstrated that values close to 1.0 can also be obtained for a nonlinear relationship [14,15]. Figure 13.1 shows calibration data that gave an r^2 close to one but the relationship is not a straight line. Therefore, visual observation of the plot of the response (y) against the concentration of the analyte or the amount spotted (x) is advised. It may be followed by plotting the y -residual values against the

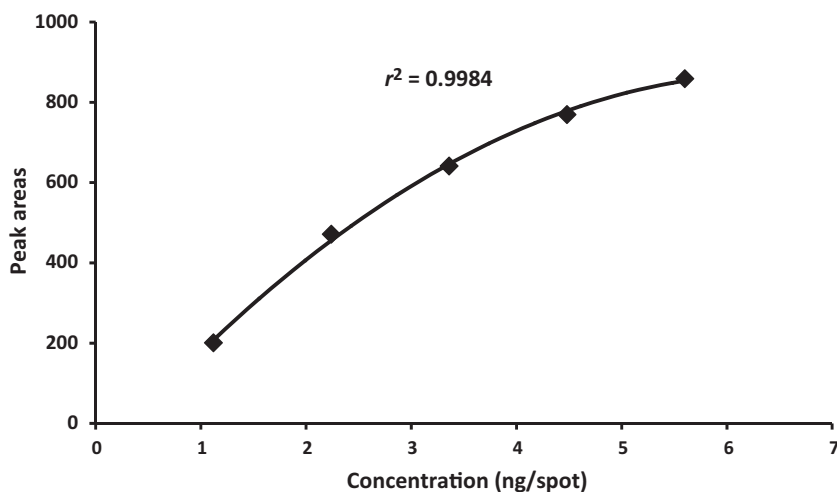


FIGURE 13.1 A nonlinear calibration curve showing a coefficient of determination (r^2) close to 1 (above 0.99).

concentration when there are enough measurements, in order to evaluate whether a trend can be seen [11,13,16]. For a straight-line situation the residuals do not show a trend, but they do when a curved behavior of the response is modeled by a straight line (i.e., by an improper model). However, statistical tests, such as a lack-of-fit (LOF) test (F-test) [16–18] or a test for the significance of the quadratic coefficient b_2 in the model $y = b_0 + b_1x + b_2x^2$ [16] are most appropriate for evaluating the linearity of the calibration line.

13.2.2 Precision (Repeatability and Time-Different Intermediate Precision)

The precision of an analytical procedure “expresses the degree of closeness of agreement (degree of scatter) between a series of measurements obtained from multiple sampling of the same homogeneous sample under the prescribed conditions” [1]. It is important to note that a precision study should always include all steps of the method from the sample preparation to the final measurements. Regulatory guidelines stipulate repeatability, intermediate precision, and reproducibility as the types of precisions possibly to be evaluated [1,2]. Precision studies for formulation assays should preferably be performed at three concentration levels with replicates at each level. Such concentration levels are 80, 100, and 120% of the expected sample concentration. In general, the number of concentration levels and their range will depend on the purpose of the method.

Repeatability expresses the precision under the same operating conditions (one instrument, one analyst) over a short interval of time, while the intermediate precision expresses within-laboratory variations possibly due to different days, different analysts, different equipment, different calibrations, or combinations of these factors. Reproducibility evaluates precision between laboratories, thus acquiring an interlaboratory study. Reproducibility is not covered in this chapter. Since analytical methods within a laboratory are used on different days and possibly by several analysts, sometimes also using different equipment, it is important that the repeatability and intermediate precision are evaluated. Literature search within published, in-house developed, and validated TLC/HPTLC methods for pharmaceutical analysis to assay active drugs in oral solid and liquid dosage forms, indicated appropriate repeatability and time-different intermediate precision values expressed as %RSD to be $\leq 3\%$ and $\leq 5\%$, respectively [19].

In practice, the intermediate precision usually is studied by measuring on several days, and occasionally varying also the analysts and/or equipment [1]. Some authors recommend a simultaneous evaluation of the repeatability and the time-different intermediate precision using an

experimental design with replicated, usually duplicated, measurements under repeatability conditions, at different days [1,17,20]. When such experimental design is employed, an analysis of variance (ANOVA) is used to estimate the individual variance components, i.e., the repeatability variance (s_r^2) and the between-days variance ($s_{\text{between-days}}^2$). Then the time-different intermediate precision ($s_{I(t)}^2$) is estimated as

$$s_{I(t)}^2 = s_r^2 + s_{\text{between-days}}^2 \quad (13.1)$$

It has been shown that the time-different intermediate precision is underestimated when the above experimental design is not used [20]. It is described that, without the experimental design, the time-different intermediate precision was estimated as %RSD = 5.2, while with the design and from the ANOVA it was estimated as %RSD = 6.0 for the same method.

13.2.3 Trueness

Trueness of an analytical method “expresses the closeness of agreement between the mean value obtained from a series of measurements and the value which is accepted either as a conventional true value or an accepted reference value” [21,22]. It is measured as recovery or bias. For a method assaying pharmaceutical formulations it is evaluated by spiking with reference standard(s) of the analytes of interest to the prepared blank matrix or to preanalyzed samples, i.e., by using a so-called standard-addition approach. The ICH guidelines [1] indicate that recovery studies for the above-mentioned assay methods are to be performed at a minimum of three concentration levels and three replicates at each level. Preferably, recoveries are determined at 80, 100, and 120% of the expected sample concentration.

13.2.4 Range

According to the ICH guidelines [1], the range of the method is “the interval between the upper and lower concentrations of the analyte in samples (including these concentrations) where methods demonstrate a suitable level of precision, trueness and linearity”. The ICH guidelines propose concentration levels covering 80–120% of the target concentration to be evaluated as the range of a particular method to assay a drug substance or a finished (drug) product. In many cases, authors report the ranges as the lower and upper concentrations levels of the calibration lines or most extremely as the upper and lower quantitation limits, which usually exceed the 80–120% of the target sample concentration. The range of the method can be easily determined when a

method is validated by the alternative approach using the accuracy profiles (see [Section 13.3](#)).

13.2.5 Detection Limit and Quantitation Limit

For an analytical method, the detection limit is “the lowest amount of analyte in a sample which can be detected but not necessarily quantitated as an exact value” and the quantitation limit is “the lowest amount of analyte in a sample which can be quantitatively determined with suitable precision and accuracy” [1]. Determination of detection limit (DL) and quantitation limit (QL) is generally not necessary for assay methods, because the above-mentioned range (80–120%) is largely within the [lower QL, upper QL] interval.

However, when they have to be estimated, several approaches are available. A first approach applies the signal-to-noise ratio (S/N) [1]. This approach is based on the measured signals (S) from sample solutions with known low concentrations and baseline signals (N) or signals from appropriate blank solutions. The S/N of 3:1 and 10:1 are generally considered acceptable for estimating the DL and the QL, respectively, as illustrated in [Figure 13.2](#).

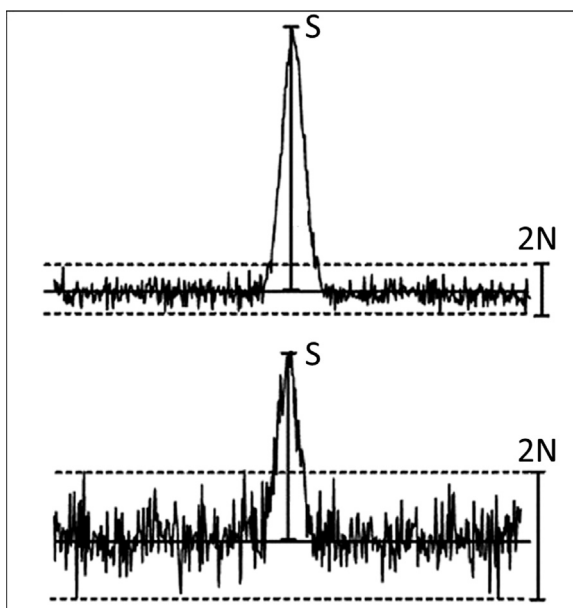


FIGURE 13.2 Simple illustration of the measurement of signal (S) and noise (N) to estimate the signal-to-noise ratio (S/N) as is used to determine DL and QL. Adapted from Ref. [31].

A second approach that can be handled to estimate the DL and the QL is using the calibration curve. According to the ICH guidelines [1], the following equations are used to compute the DL and QL:

$$DL = \frac{3.3\sigma}{S} \quad (13.2)$$

$$QL = \frac{10\sigma}{S} \quad (13.3)$$

where

σ = standard deviation of the response

S = slope of the calibration line

The slope is estimated from the calibration curve of the sample. The estimates of σ may be obtained from the standard deviation of the signal from blank solutions. The σ may also be obtained from the standard deviation of the y -intercept of regression lines prepared by solutions with sample concentrations in the range of the corresponding DL and QL. The residual standard deviation of such regression line may also be used as an estimation of σ .

A third approach used to estimate the DL is through dilution of a standard solution to the lowest detectable level [1,23].

13.2.6 Robustness

The robustness of an analytical procedure is “a measure of its capacity to remain unaffected by small, but deliberate variations in method parameters and provides an indication of its reliability during normal usage” [1]. A robustness test is performed at the end of method development/optimization in order to identify factors susceptible to cause variations in the method response. These factors then can be suitably controlled by including precautionary statements in the operating procedure for the method. Initially these influential factors were identified prior to performing an interlaboratory study to estimate the method reproducibility. These factors, when insufficiently controlled, cause the failure of laboratories within such interlaboratory studies, a fact that should be avoided [24–26].

13.2.7 Example of HPTLC Method Validated Using the Classic Approach

In this section an example is presented of an HPTLC method that was validated using the classic approach. The method was developed for the simultaneous analysis of LVD, stavudine (STV), and NVP in fixed-dose

combination tablets [27]. In this method, separation was performed on a silica gel 60 F₂₅₄ plate while the mobile phase consisted of ethyl acetate, methanol, toluene, and concentrated ammonia (12:6:12:1, v/v/v/v). The volume of each of the standard and sample solutions applied on the plate as bands were 2 μ L. Detection wavelength was set at 254 nm. The retardation factor (R_f) values after separation were 0.24, 0.38, and 0.69 for LVD, STV, and NVP, respectively.

13.2.7.1 Linearity of the Calibration Line

A stock standard solution with 524.0, 480.0, and 520.0 mg/L LVD, STV, and NVP, respectively, was prepared and diluted to five standard solutions. A volume of 2 μ L of each standard solution was applied on the HPTLC plate to deliver about 42.0, 84.0, 126.0, 168.0, and 209.0 ng LVD per spot; 38.0, 77.0, 115.0, 154.0, and 192.0 ng STV per spot; and 42.0, 83.0, 125.0, 166.0, and 208.0 ng NVP per spot. This was done in triplicate and repeated for three days. After each application of standard solutions, the plate was developed, dried, scanned, and the spot areas recorded.

For each compound, the homoscedasticity of the variances along the regression line was verified using the Cochran's criterion, C , as

$$C = \frac{s_{\max}^2}{\sum_j s_j^2} \quad (13.4)$$

where $s_{\max}^2$ and $\sum_j s_j^2$ are the highest variance and sum of all variances, respectively. C is then compared with a critical value. Since the homoscedasticity requirement was fulfilled for the three regression lines, the slope and the intercept with their 95% confidence intervals could be calculated using ordinary least squares regression. The linearity was evaluated visually by looking at the calibration curves, and statistically by performing an F -test for LOF:

$$F = \frac{MS_{\text{LOF}}}{MS_{\text{PE}}} \quad (13.5)$$

where MS_{LOF} is the mean square due to lack of fit and MS_{PE} is the mean square due to pure experimental error. The F value is compared with a critical value originating from an F -distribution.

The results showed that the equations of the calibrations lines for LVD, STV, and NVP were $\text{Area}_{\text{LVD}} = 6.29C_{\text{LVD}}(\text{ng/spot}) + 109.12$, $\text{Area}_{\text{STV}} = 5.95C_{\text{STV}}(\text{ng/spot}) + 177.79$, and $\text{Area}_{\text{NVP}} = 5.90C_{\text{NVP}}(\text{ng/spot}) + 129.90$, respectively. The corresponding values of the slopes and intercepts with their 95% confidence limits were 6.29 ± 0.08 and 109.12 ± 25.73 for LVD, 5.95 ± 0.12 and 177.79 ± 33.62 for STV, and 5.90 ± 0.10 and 129.90 ± 29.79 for NVP. The correlation coefficients, r , were 0.9998, 0.9997, and 0.9998, respectively. Visual observation of the calibration curves gave the

impression that they were linear. The LOF test results for the calibration data of LVD, STV, and NVP were $F_{\text{calc}} = 0.451, 1.390, \text{ and } 0.584$, respectively. These values were smaller than the critical value, $F_{\text{tab}}(\alpha=0.05; df_1=3, df_2=40) = 2.839$. Thus, straight lines were considered adequate to describe the relationships between the spot areas and the concentrations for each compound. It can also be noticed that these lines were not going through the origin.

13.2.7.2 Precision

The repeatability and time-different intermediate precision were determined simultaneously using an experimental design setup. To prepare the precision samples, tablet matrix powder was spiked with reference standards of LVD, STV, and NVP at 80, 100, and 120% of the target concentration of each compound. The obtained solutions were applied on the HPTLC plates to form bands with about 96, 120, and 144 ng/spot LVD; 91, 114, and 137 ng/spot STV; and 96, 120, and 144 ng/spot NVP. The analysis was done in three replicates daily and repeated for 6 days. Calibration curves to estimate the concentration of drug per spot were measured daily. The repeatability, s_r^2 , and the time-different intermediate precision, $s_{I(t)}^2$, were then estimated at each concentration level from an ANOVA table and Eqn (13.1).

The repeatability variances for LVD at the 80, 100, and 120% concentration levels were 0.43, 0.49, and 0.25 ng², while the time-different intermediate precision variances at the same levels were 2.25, 1.77, and 1.82 ng², respectively. The pooled repeatability and time-different variances expressed as %RSD were 0.62 and 1.66%, respectively. For STV, the variances were 0.24, 0.42, and 0.22 ng² for the repeatability and 1.36, 1.79, and 1.65 ng² for the time-different intermediate precision, at the concentration levels mentioned above, respectively. The corresponding pooled variances expressed as %RSD were 0.54 and 1.27%, respectively. For NVP, the repeatability variances were 1.54, 0.28, and 0.36 ng², and the time-different intermediate precision variances 2.51, 0.67, and 1.45 ng² at the earlier mentioned concentration levels, respectively. The pooled values expressed as %RSD were 0.79 and 1.21%, respectively. The precision results compared very well with other studies reported in the literature; hence, they were considered acceptable.

13.2.7.3 Trueness

The tablet matrix powder was spiked with drug components at 80, 100, and 120% of the target sample concentrations of each compound. Extraction and dilutions were performed with methanol and the amounts of each component applied on the HPTLC plate were 100, 125, and 150 ng/spot LVD; 90, 110, and 122 ng/spot STV; and 96, 120, and 144 ng/spot NVP. Solutions were prepared in triplicate and analyzed. This

procedure was repeated for three consecutive days. Calibration curves to estimate the concentration of drug per spot were measured daily. The trueness, expressed as percentage recovery, was calculated as

$$\% \text{recovery} = \frac{C_{\text{obs}}}{C_{\text{exp}}} \times 100 \quad (13.6)$$

where C_{obs} and C_{exp} are the observed and the expected concentrations (amounts) per spot, respectively.

The overall mean %recovery for each compound was calculated as

$$\text{Mean \%recovery} = \frac{R_{\text{tot}}}{n} \quad (13.7)$$

where R_{tot} is the sum of all %recoveries and n the number of observations.

The recovery results obtained for LVD at the 80, 100, and 120% concentration levels were $98.7\% \pm 1.5$, $99.4\% \pm 1.6$, and $99.4\% \pm 1.6$, respectively. The range of %recovery was 97.0–102.5%, while the mean recovery for all concentration levels was $99.2\% \pm 1.5$.

For STV, the %recoveries at the same concentration levels were $98.7\% \pm 1.8$, $98.4\% \pm 1.5$, and $98.9\% \pm 1.3$, respectively. The range of %recovery values was 96.2–100.3%. The mean recovery value covering all concentration levels was $98.6\% \pm 1.5$. The %recovery values for NVP were $99.4\% \pm 1.7$, $99.5\% \pm 1.4$, and $98.9\% \pm 2.0$, respectively. The range was 96.4–101.6% and the overall mean recovery was found to be $99.3\% \pm 1.7$. In conclusion, the method was considered to have an acceptable recovery and trueness.

13.2.7.4 Specificity

Specificity is “the ability to assess unequivocally the analyte in the presence of components which may be expected to be present and typically these might include impurities, degradants, matrix, etc.” [1]. Tablet matrix without drug components and tablet matrix spiked with drug components were extracted in methanol. The solution of tablet matrix without drug components was made to enable detection of any excipient spots with similar R_f values as the drug components. Spiking of tablets matrix was performed to make a solution with 62.5, 55.0, and 110.0 mg/L of LVD, STV, and NVP, respectively.

The chromatogram of the solution from the nonspiked tablet matrix did not show any spots. On the other hand, the chromatogram of the solution from tablet matrix spiked with the three compounds showed clear, compact, and well-separated peaks of LVD, STV, and NVP (Figure 13.3). Moreover, in Figure 13.3, no other peaks eluted besides the three active compounds. Therefore, the method was considered specific.

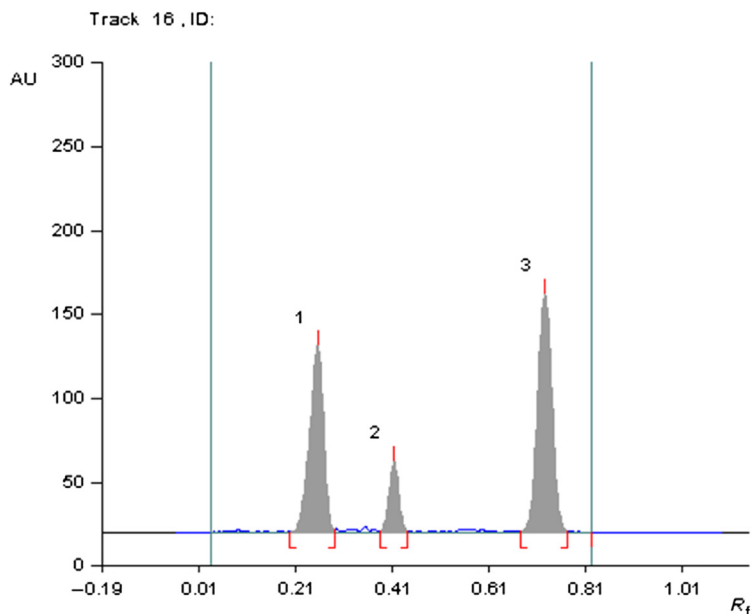


FIGURE 13.3 Chromatogram showing LVD (peak 1), STV (peak 2), and NVP (peak 3) from the solution of spiked tablet matrix. Mobile phase: ethylacetate, methanol, toluene, and concentrated ammonia (12:6:12:1, v/v/v/v). Detection at 254 nm. Reprinted with permission from Ref. [27].

13.3 ALTERNATIVE METHOD VALIDATION APPROACH USING ACCURACY PROFILES

As described earlier, TLC/HPTLC methods, or analytical methods in general, can also be validated using accuracy profiles. An accuracy profile is a single comprehensive measure of the method performance [6–8]. When constructing an accuracy profile, total errors are estimated by incorporating both systematic (absolute bias) and random errors (intermediate precision standard deviation) in order to reflect how large the measurement errors of a method can be. This approach incorporates all method performance parameters into one statistic, which enables taking decisions concerning the validity of a method for future routine use in the scope of its intended purpose.

13.3.1 Theory of Validation Using Accuracy Profiles

For the effective application of the alternative approach to the validation of TLC/HPTLC methods, a more detailed theoretical explanation of the accuracy profiles application is given below.

13.3.1.1 Protocols

The theoretical aspects of method validation using accuracy profiles are detailed in SFSTP guidelines [6–8]. These guidelines describe that, when no prior information is available regarding the characteristics of the analytical method to be validated, a prevalidation step should be performed. The information is usually gathered during the method development phase and includes, for instance, the knowledge on the presence or absence of matrix effects and the concentration levels for calibration to be used in routine application of the method. However, if such information for an analytical method is already available, one can immediately proceed to the validation step. During both prevalidation and validation, the preparation of calibration and validation standards should follow the protocol that will be implemented during routine application of the method. Calibration standards have known concentrations and can be prepared with or without matrix. Validation standards, on the other hand, are samples, reconstituted in the matrix, or any other reference material, with known concentrations, and are used to validate an analytical procedure [7]. The validation standards represent the future samples that the method will have to quantify; hence, they have to be prepared independently.

Protocols for the prevalidation or the actual validation are proposed in the guidelines for method validation using accuracy profiles [7]. In this chapter only the validation phase will be discussed. The protocols suggest a number of concentration levels for the calibration ($j = 1:c$) and validation standards ($m = 1:v$) that preferably should be used during validation, as well as the number of replicates at each concentration level of the calibration ($k = 1:d$) and validation standards ($n = 1:w$), and the number of series ($i = 1:s$), i.e., the number of replicates of the above procedure. Each protocol, therefore, has a certain number of experiments that should be performed. These experiments require independently prepared validation standards in order to obtain a good estimation of the within-series repeatability, the between-series precision, and the intermediate precision.

Five protocols are described in [7], called V1, V2, V3, V4, and V5. The choice of a given protocol depends on the presence or absence of matrix effects and on the concentration levels for calibration to be used in routine application of the method. For example, in V2, which is applicable in the absence of matrix effects, the number of concentration levels for the calibration and validation standards is $c = 2$ or 3 and $v = 3$, respectively. The number of replicates at each concentration level of the calibration and validation standards is $d = 2$ and $w = 3$, respectively. Finally, the number of series is $s = 3$. Depending on whether $c = 2$ or 3 , 39 or 45 experiments are required, respectively. Depending on the objective of the analytical

method, in each protocol the number of experiments can be changed (decreased or increased), e.g., the number of replicates at each concentration level of the validation standards can be decreased from $w = 3$ to a minimum of $w = 2$, or the number of concentration levels of the validation standards can be increased from $v = 3$ to $v = 5$.

13.3.1.2 Response Functions

After measuring the calibration standards the relationship between the response (y) and the concentration (x) should be determined [8]. Both linear and nonlinear functions can be selected to fit the calibration data. Linear functions include a straight line, with or without intercept, and a quadratic function, while the tested nonlinear functions include four-parameter and five-parameter logistics [8]. For most quantitative methods to assay active compounds in pharmaceutical formulations, a straight-line model, represented by Eqn (13.8), is preferred and adequate [8,16].

$$y = \alpha_i + \beta_i x \quad (13.8)$$

where α_i and β_i are the true intercept and the true slope of the line, respectively, which are estimated for each series $i = 1:s$ from the results of the calibration standards.

For each compound, the estimated response function $y = a + bx$ with a and b the estimated intercept and slope, respectively, for each series ($i = 1:s$), is then used to estimate the concentration (x_{mn}) of the considered compound in all validation standards (for $m = 1:v$, and $n = 1:w$) of the same series.

$$x_{mn} = \frac{y_{mn} - a}{b} \quad (13.9)$$

It should be noted that quadratic functions are not uncommon to TLC/HPTLC calibration lines [19].

13.3.1.3 Estimation of Precision

At each considered concentration level for the validation standards ($m = 1:v$), the repeatability and intermediate precision variances are estimated. To estimate these variances, ANOVA or the restricted maximum likelihood method is proposed [8]. The restricted maximum likelihood method is a particular form of maximum likelihood estimation, which uses a likelihood function calculated from a transformed set of data. In case of variance estimation, the original data set is replaced by a set of contrasts calculated from the data and the likelihood function is calculated from the probability distribution of these contrasts, according to the model for the complete data set. The restricted maximum likelihood

method can be used in situations where some validation standard data are missing, therefore making the repeated data within concentration levels to be unbalanced. In such conditions, the variance estimation using, for example, ANOVA, would be biased [8].

However, because the validation procedure is normally well planned, the number of validation standards at each concentration level ($n = 1:w$) is usually balanced. In such a situation, the different variance components can be estimated using ANOVA [16]. For each concentration level of the validation standards ($m = 1:v$), the repeatability or intraseries variance, $s_{r,m}^2$, is estimated. The time-different intermediate-precision variance, $s_{I(t),m}^2$, is then determined as follows (Eqn (13.10)):

$$s_{I(t),m}^2 = s_{r,m}^2 + s_{\text{between-days},m}^2 \quad (13.10)$$

where $s_{\text{between-days},m}^2$ represents the between-series variance. These precision estimates are then expressed as %RSD.

13.3.1.4 Estimation of Trueness

The trueness of an analytical method expresses the closeness of agreement between the average of a series of measurements $\bar{x}_m$ and the accepted reference value $\hat{\mu}_m$ [7,8,21,22], and estimates the systematic error of an analytical method. At each concentration level of the validation standards ($m = 1:v$), trueness can be expressed as bias_m (Eqn (13.11)), absolute bias_m (Eqn (13.12)), %bias_m (Eqn (13.13)), or as %recovery (Eqn (13.14)) [8,16]:

$$\text{bias}_m = \bar{x}_m - \hat{\mu}_m \quad (13.11)$$

$$|\text{bias}_m| = |\bar{x}_m - \hat{\mu}_m| \quad (13.12)$$

$$\% \text{bias}_m = \frac{\bar{x}_m - \hat{\mu}_m}{\hat{\mu}_m} \times 100 \quad (13.13)$$

$$\% \text{recovery}_m = \frac{\bar{x}_m}{\hat{\mu}_m} \times 100 \quad (13.14)$$

where $\bar{x}_m$ is the average of the back-calculated concentrations, and $\hat{\mu}_m$ is the average of the introduced amounts of the validation standards at each concentration level ($m = 1:v$).

13.3.1.5 Estimation of Accuracy

The accuracy of an analytical method expresses the closeness of agreement between the trial result x_{imn} and the accepted reference value μ_{imn} for each measured validation standard ($i = 1:s$, $m = 1:v$, $n = 1:w$) [8,16].

It includes both the trueness and precision of the method. Accuracy (Eqn (13.15)) and %accuracy (Eqn (13.16)) are estimated as follows:

$$\text{accuracy}_{imn} = x_{imn} - \mu_{imn} \quad (13.15)$$

$$\% \text{accuracy}_{imn} = \frac{x_{imn} - \mu_{imn}}{\mu_{imn}} \times 100 \quad (13.16)$$

13.3.1.6 Total Error

Each individual measurement, x_{imn} ($i = 1:s, m = 1:w, n = 1:w$), is the sum of the true value, μ_{imn} , the absolute bias, $|\text{bias}_m|$, and the intermediate precision standard deviation $s_{I(t),m}$:

$$\begin{aligned} x_{imn} &= \mu_{imn} + |\text{bias}_m| + s_{I(t),m} \\ \Leftrightarrow x_{imn} - \mu_{imn} &= |\text{bias}_m| + s_{I(t),m} = \text{absolute total error} \end{aligned} \quad (13.17)$$

The absolute total error of an analytical method thus is the sum of the trueness ($|\text{bias}_m|$) and precision ($s_{I(t),m}$) contributions, and indicates the ability of the analytical method to produce accurate results [8]. It enables the assessment of the quantitative performance of a method. The total error estimation is an essential step in the assessment of the validity of an analytical method, as it is used to calculate the 95% tolerance interval, which provides guarantee of what results will be produced by the method in the future, during routine use of the method [8].

13.3.1.7 Tolerance Interval and Accuracy Profile

The parameters $\% \text{bias}_m$, $s_{r,m}^2$, $s_{\text{between-days},m}^2$ and $\% \text{RSD}_{I(t),m}$ are then used to calculate the expected proportion of observations that falls within the predefined bias acceptance limits $[-\lambda, +\lambda]$ (Figure 13.4). This expected proportion is calculated using the two-sided 95% β -expectation tolerance interval at each concentration level of the validation standards ($m = 1:w$). The lower L_m and the upper U_m tolerance interval limits are determined as follows:

$$L_m = \% \text{bias}_m - Q_{t(v, \frac{1+\beta}{2})} \sqrt{1 + \frac{1}{swB_m^2}} \% \text{RSD}_{I(t),m} \quad (13.18)$$

$$U_m = \% \text{bias}_m + Q_{t(v, \frac{1+\beta}{2})} \sqrt{1 + \frac{1}{swB_m^2}} \% \text{RSD}_{I(t),m} \quad (13.19)$$

where $Q_{t(v, \frac{1+\beta}{2})} = \beta$ quantile of the Student t distribution with ν degrees of freedom, with $\nu = \frac{(R_m+1)^2}{\frac{(R_m+1/w)^2}{(s-1)} + \frac{(1-1/w)}{sw}}$ where s is the number of series, w the

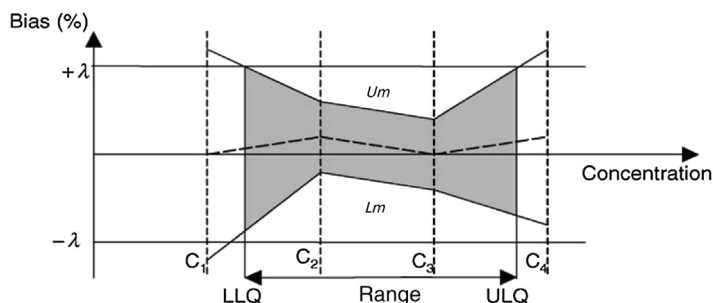


FIGURE 13.4 Accuracy profile showing the acceptance limits ($+\lambda$, $-\lambda$), the upper and the lower tolerance limits (continuous lines), and the intersection points that define the upper and lower quantification limits (ULQ and LLQ). C1–C4: concentration levels. Reprinted with permission from Ref. [30].

number of replicates for the validation standards, and $B_m^2 = \frac{R_m + 1}{wR_m + 1}$ with $R_m = \frac{s_{\text{between-days},m}^2}{s_{r,m}^2}$.

After calculating the two-sided 95% β -expectation tolerance limits at each concentration level of the validation standards ($m = 1:v$), the resulting L_m , U_m , $-\lambda$, and $+\lambda$ at the different concentration levels are then connected to construct the accuracy profile [7,8,20] (Figure 13.4). An accuracy profile allows visually observing the ability of an analytical method to fulfill its objectives and to control the risks associated with its routine application [6,7]. When the 95% β -expectation tolerance interval limits for the considered concentration levels of the validation standards fall within the bias acceptance limits, it is guaranteed that future results of the method will fall within the acceptable bias limits.

The accuracy profile reflects the adequacy of the response function used to calculate concentrations of the validation standards. In addition, the upper and the lower concentrations of the validation standards, considered during the validation study, define the upper and the lower quantification limits. In case the tolerance limits intersect with the bias acceptance limits, the intersection points will define the upper and lower quantification limits (Figure 13.4).

13.3.1.8 Measurement and Expanded Measurement Uncertainties

The measurement uncertainty is a parameter that characterizes the dispersion of the values that could reasonably be attributed to the measured compound [28]. It is used to indicate whether measurement result complies with specifications, such as regulatory or manufacturing limits [29]. The measurement uncertainty, $u(Y)$, is calculated at each

concentration level of each compound, according to Eqns (13.20), (13.21), and (13.22) [28]:

$$u(Y)^2 = s_{I(t)}^2 + u(\hat{\delta})^2 \quad (13.20)$$

with $s_{I(t)}$ the time-different intermediate precision, and $u(\hat{\delta})$ the uncertainty associated with the bias δ of the method. The latter is derived as follows:

$$u(\hat{\delta})^2 = \frac{s_{I(t)}^2 (1 - \gamma + \frac{\gamma}{w})}{s} \quad (13.21)$$

with $\gamma = \frac{s_r^2}{s_{I(t)}^2}$, w the number of replicates, and s the number of series of the validation standards.

The measurement uncertainty, $u(Y)$, is then used to derive the expanded measurement uncertainty $U(Y)$ as follows:

$$U(Y) = 2u(Y) \quad (13.22)$$

where 2 is the coverage factor based on the 95% confidence level.

The expanded uncertainty provides an interval within which the value of the measurand is believed to lie with the above level of confidence.

13.3.2 Example of an HPTLC Method Validated Using Accuracy Profiles

A normal-phase HPTLC method for the analysis of sulfamethoxazole (SMX) and trimethoprim (TMP) in co-trimoxazole tablets was validated using accuracy profiles [30]. Validation protocol V2 was found to be the most appropriate. Calibration standard solutions were prepared without tablet matrix, while the validation standards were prepared independently in the tablet matrix. The number of series in V2 is three, thus this protocol requires at least 45 experiments. The low, intermediate, and high concentration levels for both calibration and validation standards were 80, 100, and 120% of the target concentration of each compound in the tablet formulation. For both calibration and validation standards the concentrations were 120, 150, and 180 ng/spot band for SMX and 150, 180, and 216 ng/spot band for TMP. Duplicate measurements were made for the calibration standards and triplicate for the validation standards at each concentration level, daily for 6 days. The calibration and validation standards were spotted on the HPTLC plates as bands. The calibration-standard amounts at each concentration level (x_{mn}) were also estimated using (Eqn (13.9)) and expressed as nanograms per band, i.e., 119.6, 149.4, and 181.1 for SMX and 150.2, 179.2 and 215.8 for TMP.

The straight-line models were evaluated by an F-test for LOF at 5% significance level and were found to be adequate (see [Section 13.2.7.1](#)). Therefore, a straight-line response function was built for both SMX and TMP. Then, the amount of each compound in the validation standards was estimated using [Eqn \(13.9\)](#).

Since the validation data were balanced, ANOVA instead of the maximum likelihood method was used to estimate the repeatability and time-different intermediate precision variances ([Eqn \(13.10\)](#)). For both SMX and TMP the %RSD values for repeatability and time-different intermediate precision did not exceed 1.63 and 1.95%, respectively. The repeatability, the between-days variances, and the %RSD of the time-different intermediate precision for each compound at each concentration level are indicated in [Table 13.1](#), and were used to estimate the 95% tolerance intervals.

Trueness was expressed as %bias or %recovery. The %bias values calculated using [Eqn \(13.13\)](#) at each of the three concentration levels were -0.35 , -0.42 , and 0.62 for SMX; and 0.31 , -0.46 , and -0.08 for TMP. These values are much lower than the limits set for active ingredients in pharmaceutical formulations, i.e., $\pm 5\%$, and indicate a very low bias and the absence of matrix effects. The %bias values are also used in the computation of the 95% tolerance interval. Trueness was also presented as %recovery ([Eqn \(13.14\)](#)) and the obtained values ranged from 99.0 to 100.6% for all compounds.

The accuracy and %accuracy for SMX and TMP were estimated using [Eqns \(13.15\) and \(13.16\)](#). The highest values of %accuracy for SMX and

TABLE 13.1 Precision Results: Repeatability, Between-Days Variances, and Time-Different Intermediate Precision for SMX and TMP at Each Concentration Level ($w = 3$, $v = 3$, $s = 6$)

Compound	Concentration Level (%)	Repeatability			Time-Different Intermediate Precision	
		Standard Deviation (ng/spot)	% RSD	Between-Days Variance	Standard Deviation (ng/spot)	% RSD
SMX	80	0.864	0.72	2.074	1.680	1.40
	100	1.274	0.85	0.418	1.429	0.95
	120	0.766	0.42	0.165	0.867	0.48
TMP	80	2.456	1.63	2.613	2.940	1.95
	100	1.739	0.97	4.895	2.814	1.57
	120	1.663	0.77	8.232	3.316	1.54

TMP were 2.18 and 3.56, respectively, and both were well within the bias limits of $[-5\%, +5\%]$.

The absolute total error values were calculated using (Eqn (13.17)) at each concentration level. The values (in nanogram per band) obtained for SMX were 2.1, 2.1, and 2.0, whereas for TMP they were 3.4, 3.6, and 3.5. The values of the total error are also used in calculating the 95% tolerance intervals of the methods.

For each compound, the bias acceptance limits for the total errors were selected to be $\pm 5\%$, which are often used for active ingredients in pharmaceutical formulations. The lower L_m and the upper U_m tolerance interval limits for each concentration level were calculated using Eqn (13.18) and (Eqn (13.19)), respectively. Table 13.2 shows the two-sided 95% β -tolerance limits for each compound at every concentration level studied. The accuracy profiles of each compound are presented in Figure 13.5.

These accuracy profiles show that the tolerance limits for all compounds fall well within the bias acceptance limits set at $\pm 5\%$. This means that the HPTLC method is capable of obtaining accurate results at the studied concentration ranges (120.0–180.0 ng/band for SMX and 150.0–216.0 ng/band for TMP). The risk of having future measurements falling outside the $\pm 5\%$ acceptance limits at the studied concentration levels is strictly controlled; hence, the method is valid to be applied in the routine assay of the compounds in the co-trimoxazole fixed-dose combination tablets.

Furthermore, this accuracy profile approach also confirms the selection of the straight-line response functions for the quantification of SMX and TMP in their tablets. The lowest and the highest concentrations studied for each compound (as indicated in the range above) can be considered as the lower and upper QLS, respectively. Most probably still lower and higher limits can be defined, but such values were not validated for the actual methods. Within the actual quantification limits, the measurement results have acceptable accuracy because their 95% tolerance limits do not

TABLE 13.2 The Two-Sided 95% Tolerance Limits for Each Compound at Each Concentration Level

Compound	Concentration (ng/spot)	95% Tolerance Limits
SMX	119.6	$[-3.7; 3.0]$
	149.4	$[-2.5; 1.6]$
	181.1	$[-0.4; 1.6]$
TMP	150.5	$[-4.4; 5.0]$
	179.2	$[-4.0; 3.1]$
	215.8	$[-3.7; 3.5]$

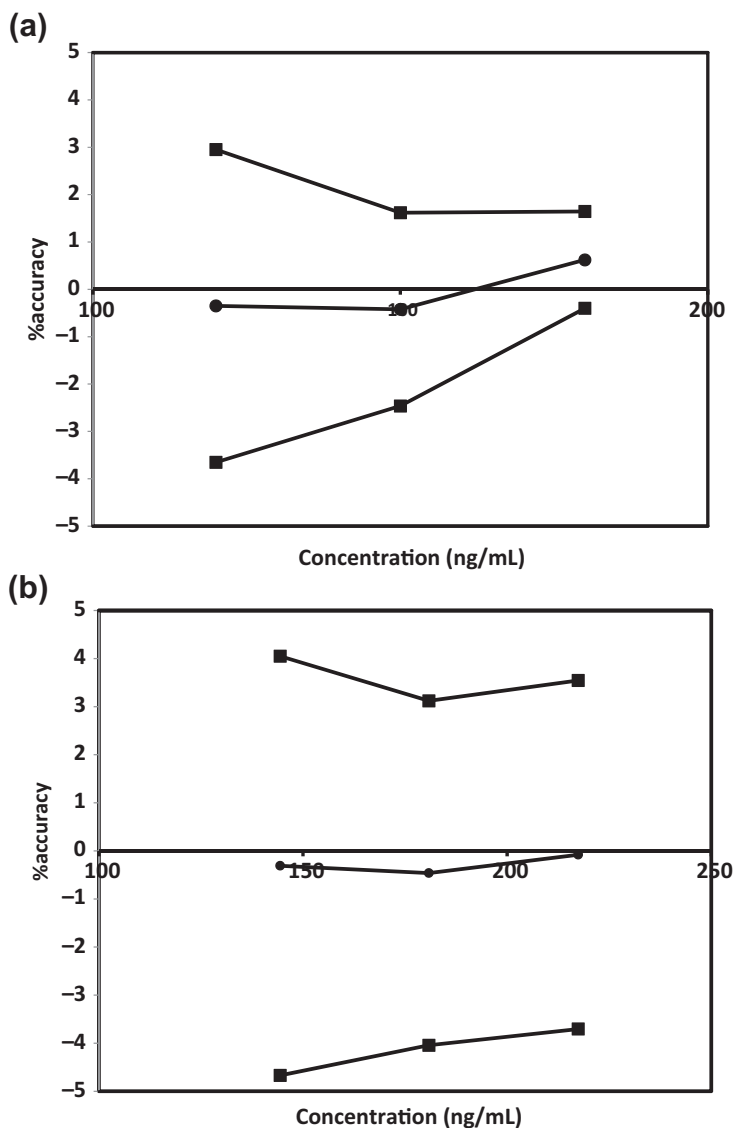


FIGURE 13.5 The accuracy profiles of (a) SMX and (b) TMP. Legend: ● %bias, ■ tolerance limit.

exceed the $\pm 5\%$ acceptable limits. Therefore, for each compound the interval from the lowest to the highest tested concentration can be regarded as the range of the corresponding method.

The measurement uncertainties and expanded measurement uncertainties (nanograms per band) for each compound at each concentration

TABLE 13.3 Measurement Uncertainty, $u(Y)$ (ng/spot), and Expanded Uncertainty, $U(Y)$ (ng/spot), for Each Compound at Each Concentration Level

Compound	Concentration (ng/spot)	Measurement Uncertainty $u(Y)$ (ng/spot)	Expanded Uncertainty $U(Y)$ (ng/spot)
SMX	119.6	1.8	3.6
	149.4	1.5	3.0
	181.1	0.9	1.8
TMP	150.5	3.1	6.1
	179.2	3.0	6.0
	215.8	3.5	7.1

level of SMX and TMP are presented in Table 13.3. The highest and the lowest values indicate the range of uncertainty over the concentration levels of each compound, and are used to determine compliance of results to specifications during routine use of the validated method. In determining the compliance of a sample characteristic to the set specifications, the measured result with the measurement uncertainty value are compared to the upper and lower value of the specification. For instance, suppose 40 ng/band is the measured result, 2 ng/band the measurement uncertainty of the method, and 35–45 ng/band the set specification. The result is spread as 40 ± 2 ng/band (i.e., 38–42 ng/band), therefore complying with the specifications. For further description of the compliance assessment using measurement uncertainty, the reader is referred to reference [29].

13.4 CONCLUSION

Analytical method validation is as important for quantitative TLC/HPTLC methods as for any other analytical method. Both the classic and alternative method validation approaches, the latter using accuracy profiles, can be used to validate these methods.

When properly optimized and validated, quantitative TLC/HPTLC methods result in adequate assays.

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